





**Barker**  
College

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Student Number

**2023**

**YEAR 12  
ASSESSMENT BLOCK**

# Mathematics Extension 2

**Staff Involved:**

BHC (Mr Carruthers)  
DZP (Mr Peattie)  
ARM\* (Mr Mallam)

**12:00 NOON  
FRIDAY 31 MARCH**

**35 copies**

**TOPICS COVERED:**

Complex Numbers I & II

Proof

## General Instructions

- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale
- Write your student number at the top of this page

**Total marks:  
80**

**Section I – 8 marks** (pages 4-6)

- Attempt Questions 1-8
- Allow about 15 minutes for this section

**Section II – 72 marks** (pages 7-12)

- Attempt Questions 9-14
- Allow about 1 hour and 45 minutes for this section

## Section I

**8 marks**

**Attempt Questions 1-8**

**Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1-8.

**1** What is the negation of the statement: “If it is raining, then I carry an umbrella”?

- A. If it is not raining, then I do not carry an umbrella.
- B. If I am carrying an umbrella, then it is raining.
- C. It is not raining and I am carrying an umbrella.
- D. It is raining and I am not carrying an umbrella.

**2** What is the value of  $(4 - 3i)^2$ ?

- A.  $7 - 24i$
- B.  $7 + 24i$
- C.  $25 - 24i$
- D.  $25 + 24i$

**3** If  $n$  is a positive integer, for how many values of  $n$  is  $9n^2 - 4$  a prime number?

- A. 0
- B. 1
- C. 2
- D. More than 2

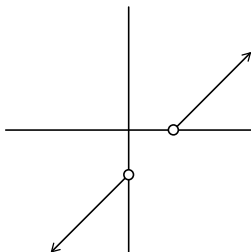
- 4 Which of the following best describes the locus of points in the Argand plane given by

$$|z - 2| = 2|z + 1|?$$

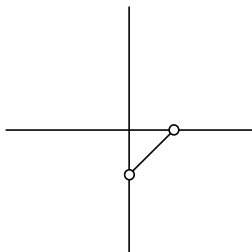
- A. A vertical line through (0, 0).
- B. A vertical line through (1, 0).
- C. A circle with centre (0, 0).
- D. A circle with centre (-2, 0).

- 5 Which diagram best represents the solutions to the equation  $\text{Arg}(z - 2) = \text{Arg}(z + 2i)$ ?

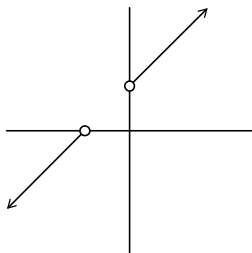
A.



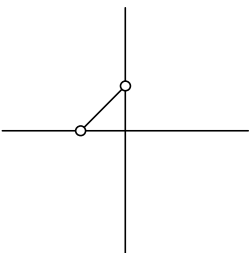
B.



C.



D.



- 6 For complex  $z$ , if  $\text{Im}(z) > 0$ , what is the value of  $\text{Arg}\left(\frac{z\bar{z}}{z - \bar{z}}\right)$ ?

A.  $-\frac{\pi}{2}$

B. 0

C.  $\frac{\pi}{2}$

D.  $\pi$

- 7 For integers  $x$  and  $y$ , which one of the following statements is equivalent to the statement: "There is no integer which is the largest integer"?

A.  $\exists y (\forall x, y > x)$

B.  $\forall x (\exists y, y > x)$

C.  $\exists x (\forall y, y > x)$

D.  $\forall y (\exists x, y > x)$

- 8 Let  $(x + iy)^{18} = a + ib$  for  $a, b, x, y \in \mathbb{R}$ . Which expression is equivalent to  $(y - ix)^{18}$ ?

A.  $-a - ib$

B.  $b - ia$

C.  $-b + ia$

D.  $-a + ib$

**End of Section I**

## Section II

72 marks

Attempt Questions 9–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

**Question 9** (12 marks) Use a SEPARATE writing booklet.

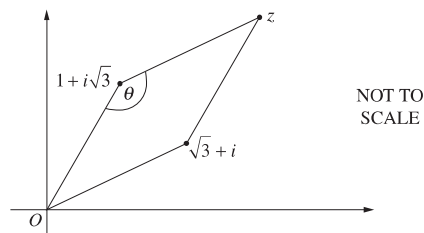
(a) Consider the complex numbers  $w = 2 - 3i$  and  $z = 3 + 4i$ .

(i) Find  $\bar{w} + z$ . 1

(ii) Find  $|w|$ . 1

(iii) Express  $\frac{w}{z}$  in the simplest form of  $x + iy$  where  $x$  and  $y$  are real. 2

(b) On the Argand diagram below, the complex numbers  $0$ ,  $1 + i\sqrt{3}$ ,  $\sqrt{3} + i$  and  $z$  form a rhombus.



(i) Find  $z$  in the form  $x + iy$ . 1

(ii) An interior angle of the rhombus,  $\theta$ , is marked on the diagram. Find the value of  $\theta$ . 2

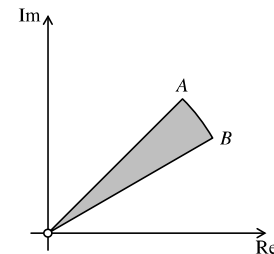
(c) Use proof by contradiction to show that  $\log_2 7$  is irrational. 2

(d) Prove that the sum of two consecutive positive even powers of 3 is always a multiple of 90. 3

**Question 10** (12 marks) Use a SEPARATE writing booklet.

(a) The diagram below shows the region on the complex plane given by:

$$|z| \leq 4\sqrt{2} \text{ and } \frac{\pi}{6} \leq \text{Arg}(z) \leq \frac{\pi}{4}.$$



(i) Find the complex number represented by point  $A$  in modulus-argument form. 1

(ii) Find the complex number represented by point  $B$  in the form  $x + iy$ . 1

(iii) Determine whether  $5 + 2i$  lies within the region. 2

(b) (i) Given that  $\cos \theta + i \sin \theta = e^{i\theta}$ , prove de Moivre's theorem: 1

for positive integers  $n$ ,  $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$ .

(ii) Using de Moivre's theorem, or otherwise, show that, for positive integers  $n$ , 2

$$(1 + i)^n + (1 - i)^n = 2 \left( \sqrt{2} \right)^n \cos \frac{n\pi}{4}.$$

(c) Two of the zeros of the polynomial  $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$  are  $a + ib$  and  $a + 2ib$ , where  $a$  and  $b$  are real and  $b > 0$ .

(i) Find the values of  $a$  and  $b$ . 3

(ii) Hence, or otherwise, express  $P(x)$  as a product of quadratic factors with real coefficients. 2

**Question 11** (12 marks) Use a SEPARATE writing booklet.

- (a) Consider the following statement for integers  $m$  and  $n$ .

$P$ : If  $\frac{mn}{2}$  is an integer then  $m$  or  $n$  must be even.

- (i) Write down the contrapositive of  $P$ . 1
- (ii) Prove  $P$  by proving the contrapositive. 3
- (b) Use mathematical induction to prove the following statement for integers  $n \geq 1$ : 4

$$\sum_{r=1}^n (r-1)^3 \leq \frac{n^4}{4}.$$

- (c) On the complex plane, the line given by  $|z+1| = |z-2+3i|$  cuts the circle given by  $|z+1| = 3$  into two segments. Find the exact area of the major segment. 4

**Question 12** (12 marks) Use a SEPARATE writing booklet.

- (a) For any integer  $n$ , prove that  $n^2 + n$  is even. 2

- (b) (i) Expand and simplify  $(\cos \theta + i \sin \theta)^5$  using the binomial theorem. 2

- (ii) Hence, using de Moivre's theorem, show that 2

$$\sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta.$$

- (iii) Show that  $x = \sin \frac{\pi}{10}$  is one of the solutions to  $16x^5 - 20x^3 + 5x - 1 = 0$ . 1

- (iv) Write down the polynomial  $P(x)$  such that  $(x-1)P(x) = 16x^5 - 20x^3 + 5x - 1$ . 1

- (v) Find the value of  $a$  such that  $P(x) = (4x^2 + ax - 1)^2$ . 1

- (vi) Hence find the exact value of  $\sin \frac{\pi}{10}$ . 1

- (vii) Sketch the function  $f(x) = 16x^5 - 20x^3 + 5x - 1$ , showing all intercepts. 2

**Question 13** (12 marks) Use a SEPARATE writing booklet.

(a)  $z = 2e^{\frac{\pi}{12}i}$  is a root of the equation  $z^4 = a(1 + \sqrt{3}i)$ , where  $a$  is real.

(i) Express  $1 + \sqrt{3}i$  in the form  $re^{i\theta}$ . 1

(ii) Find the value of  $a$ . 1

(iii) Find the other three roots of the equation in the form  $re^{i\theta}$ . 2

(b) The point  $P$  on an Argand diagram represents the complex number  $z$ , which satisfies 4

$$\frac{1}{z} + \frac{1}{\bar{z}} = 1.$$

Give a geometrical description of the locus of  $P$  as  $z$  varies.

(c) For real numbers  $a, b > 0$  where  $ab \geq 1$ , prove that 4

$$\frac{1}{a^2 + 1} + \frac{1}{b^2 + 1} \geq \frac{2}{ab + 1}.$$

**Question 14** (12 marks) Use a SEPARATE writing booklet.

(a) If  $z$  is one of the fifth roots of unity other than 1, show that 3

$$z - z^2 + z^3 - z^4 = 2i \left( \sin \frac{2\pi}{5} - \sin \frac{\pi}{5} \right).$$

(b) (i) For any integer  $a \geq 2$ , show that  $\frac{1}{a^2} < \frac{1}{a-1} - \frac{1}{a}$ . 2

(ii) Hence, for any positive integer  $n$ , show that 3

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} < 2.$$

*Suggestion: do not use induction.*

(c) (i) By considering  $f'(x)$  where  $f(x) = e^x - x$ , show that  $e^x > x$  for  $x \geq 0$ . 1

(ii) Hence, by considering areas under the curves  $y = e^x$  and  $y = x$ ,  
prove that  $\sqrt{e} > \frac{9}{8}$ . 1

(iii) Use mathematical induction and the result in (i) to show that, for  $x \geq 0$ ,  
 $e^x > \frac{x^n}{n!}$  for all integers  $n \geq 1$ . 2

**End of paper**

Q1/  $\sim(p \Rightarrow q) \equiv p \wedge \sim q$   
(D)

Q2/  $(4-3i)^2$   
 $= 16 - 24i + 9$   
 $= 7 - 24i$  (A)

Q3/  $9n^2 - 4 = (3n-2)(3n+2)$

When  $n=1$ ,  
 $RHS = 1 \times 5 = 5$  (prime).

For all other  $n$ ,  
 $RHS = a \times b$  where  $a, b > 1$

$\therefore$  no more primes

(B)

Q4/  $|z-2| = 2|z+1|$

$(x-2)^2 + y^2 = 4(x+1)^2 + 4y^2$   
 $x^2 - 4x + 4 + y^2 = 4x^2 + 8x + 4 + 4y^2$

$0 = 3x^2 + 12x + 3y^2$

$0 = x^2 + 4x + y^2$

$4 = (x+2)^2 + y^2$

(D)

Q5/

(A)

Yr 12 Ext 2  
 2023  
 Assessment 2  
 Solutions  
 31/3/2023

Q6/ Let  $z = x + iy$  then

$\text{Arg}(z\bar{z}) - \text{Arg}(z - \bar{z})$

$= \text{Arg}(x^2 + y^2) - \text{Arg}(2iy)$

$= 0 - \frac{\pi}{2}$

$= -\frac{\pi}{2}$  (A)

or  $\text{Arg } z + \text{Arg } \bar{z} = 0$   
 $\text{Arg}(z - \bar{z})$  is  $\frac{\pi}{2}$   
 using vectors.

Q7/ (B)

Q8/  $(x+iy)^{18} = a + ib$

Now  $(y-ix)$

$= \left[ i^3 \left( x + \frac{y}{i^3} \right) \right]^{18}$

$= i^{54} \left( x + \frac{y}{-i} \right)^{18}$

$= i^{54} (x + iy)^{18}$

$= (i^4)^{13} \times i^2 \times (a + ib)$

$= 1 \times -1 \times (a + ib)$

$= -a - ib$

(A)

or  $(y-ix)^{18} = \left[ -i(x+iy) \right]^{18} = i^{18} (x+iy)^{18}$

$= (i^4)^9 \cdot i^2 (x+iy)^{18}$

$= (1)(-1)(x+iy)^{18}$

$= -(x+iy)^{18}$

$= -(a+ib)$

$= -a - ib$

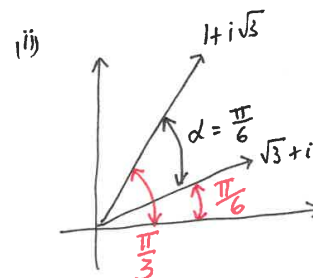
Q9/  $w = 2-3i$   
 $\bar{z} = 3+4i$

(a) (i)  $\bar{w} + z$   
 $= 2+3i + 3+4i$   
 $= 5+7i$

(ii)  $|w| = \sqrt{4+9}$   
 $= \sqrt{13}$

(iii)  $\frac{w}{z}$   
 $= \frac{2-3i}{3+4i} \times \frac{3-4i}{3-4i}$   
 $= \frac{6-8i-9i-12}{9+16}$   
 $= \frac{-6-17i}{25}$

(b) (i)  $z = (1+\sqrt{3}i) + (\sqrt{3}+i)$   
 $= (\sqrt{3}+1) + i(1+\sqrt{3})$



$\theta = \pi - \frac{\pi}{6}$  (co-interior angles are supplementary if parallel lines)  
 $\theta = \frac{5\pi}{6}$

(c) Proof by Contradiction.

Suppose that  $\log_2 7$  is rational.

Let  $\log_2 7 = \frac{p}{q}$  where  $p$  and  $q$

are integers.

Then  $7 = 2^{\frac{p}{q}}$

so  $7^q = 2^p$ .

Contradiction!

The LHS is odd while the RHS is even.

$\therefore \log_2 7$  is irrational.

(d) Let the consecutive even powers of 3 be

$3^{2n}$  and  $3^{2n+2}$ ,  $n$  integer  $\geq 1$

Then  $3^{2n} + 3^{2n+2}$

$= 3^{2n} (1 + 3^2)$

$= 10 \times 3^{2n}$

$= 10 \times 9 \times 3^{2n-2}$

$= 90 \times 3^{2n-2}$

which is a multiple of 90.  
 if  $n$  integer  $\geq 1$

Q10, Ca)

(i)  $A = 4\sqrt{2} \operatorname{cis}(\frac{\pi}{4})$

(ii)  $B = 4\sqrt{2} \operatorname{cis}(\frac{\pi}{6})$   
 $= 4\sqrt{2} (\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$   
 $= 4\sqrt{2} (\frac{\sqrt{3}}{2} + i \frac{1}{2})$   
 $= 2\sqrt{6} + 2\sqrt{2}i$

(iii) Now  $|5+2i| = \sqrt{29}$   
 and  $\sqrt{29} < 4\sqrt{2}$  ( $\approx 5.39$ )  
 it's close enough and a maybe!  
 But  $\arg(5+2i) = 21^\circ 48' 5''$  so

$(5+2i)$  does not lie in the shaded region because its argument is too small.

(b) (i) Given  $\cos \theta + i \sin \theta = e^{i\theta}$   
 prove that  $(\cos \theta + i \sin \theta)^n$   
 $= \cos(n\theta) + i \sin(n\theta)$

Now  $(\cos \theta + i \sin \theta)^n$   
 $= (e^{i\theta})^n$   
 $= e^{in\theta} = e^{i(n\theta)}$   
 $= \cos(n\theta) + i \sin(n\theta)$   
 as required.

b)(ii) Required to show that

$$(1+i)^n + (1-i)^n = 2(\sqrt{2})^n \cos \frac{n\pi}{4}$$

Now  $(1+i)^n + (1-i)^n$   
 $= (\sqrt{2} e^{i\frac{\pi}{4}})^n + (\sqrt{2} e^{-i\frac{\pi}{4}})^n$   
 $= (\sqrt{2})^n (\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4}) + (\sqrt{2})^n (\cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4})$   
 $= (\sqrt{2})^n \cos \frac{n\pi}{4} + (\sqrt{2})^n \cos \frac{n\pi}{4}$   
 $= 2(\sqrt{2})^n \cos \frac{n\pi}{4}$ , as required.

(c) All the coefficients are Real, so roots occur in conjugate pairs.

(i) The roots are  $a+ib, a-ib, a+2ib, a-2ib$  where  $a, b$  real

$$\Sigma(x) = -\frac{b}{a} = 4 = 4a \quad \therefore a = 1$$

$$\Pi(x) = (a^2+b^2)(a^2+4b^2) = \frac{c}{a} = 10$$

$$\therefore (1+b^2)(1+4b^2) = 10$$

$$1+5b^2+4b^4 = 10 \quad \text{Let } b^2 = m.$$

$$4m^2+5m-9 = 0$$

$$(4m+9)(m-1) = 0$$

$$m = -\frac{9}{4} \quad m = 1$$

$$\therefore b^2 = -\frac{9}{4} \quad \therefore b^2 = 1$$

But  $b$  real  $\therefore b = \pm 1$

$$a = 1 \text{ and } b = \pm 1$$

(ii)  $\therefore P(x) = (x-(1+i))(x-(1-i))(x-(1+2i))(x-(1-2i))$   
 $= (x^2-2x+2)(x^2-2x+5)$   
 or zeros:  $1+i, 1-i$  |  $1+2i, 1-2i$   
 Sum = 2 | Sum = 2  
 Product = 2 | Product = 5

Q11 The statement P is of the form

(i)  $(\frac{mn}{2} \text{ is an integer}) \Rightarrow (m \text{ or } n \text{ is even})$

The contrapositive is of the form

$$\sim(m \text{ or } n \text{ is even}) \Rightarrow \sim(\frac{mn}{2} \text{ is an integer}).$$

In words, the contrapositive is:

"If neither  $m$  nor  $n$  is even, then  $\frac{mn}{2}$  is not an integer."

Or: "If  $m$  and  $n$  are both odd, then  $\frac{mn}{2}$  is not an integer."

(ii) Let  $m$  and  $n$  both be odd.

$$\text{Let } m = 2j+1 \text{ and } n = 2k+1.$$

$$\text{Then } \frac{mn}{2} = \frac{(2j+1)(2k+1)}{2}$$

$$= \frac{4jk+2j+2k+1}{2}$$

$$= 2jk+j+k+\frac{1}{2}$$

Since  $j$  and  $k$  are integers,

$2jk+j+k$  is an integer.

Hence  $2jk+j+k+\frac{1}{2}$  is not an integer.

If both  $m$  and  $n$  are odd, then  $\frac{mn}{2}$  is not an integer.

$\therefore$  If  $\frac{mn}{2}$  is an integer, then  $m$  or  $n$  must be even.

(b) Prove that  $\sum_{r=1}^n (r-1)^3 \leq \frac{n^4}{4}$  for integers  $n \geq 1$ .

when  $n=1$ ,  $LHS = \sum_{r=1}^1 (1-1)^3$  and  $RHS = \frac{1^4}{4}$   
 $= 0$   $= \frac{1}{4}$

Hence  $LHS \leq RHS$  when  $n=1$ .

The statement is true when  $n=1$ .

Let the statement be true when  $n=k$ , i.e., let

$$\sum_{r=1}^k (r-1)^3 \leq \frac{k^4}{4}$$

and then try to prove that

$$\sum_{r=1}^{k+1} (r-1)^3 \leq \frac{(k+1)^4}{4}$$

Now  $LHS = \sum_{r=1}^{k+1} (r-1)^3$

$$= \sum_{r=1}^k (r-1)^3 + [(k+1)-1]^3$$

$$\leq \frac{k^4}{4} + k^3, \text{ by our assumption}$$

$$= \frac{k^4 + 4k^3}{4}$$

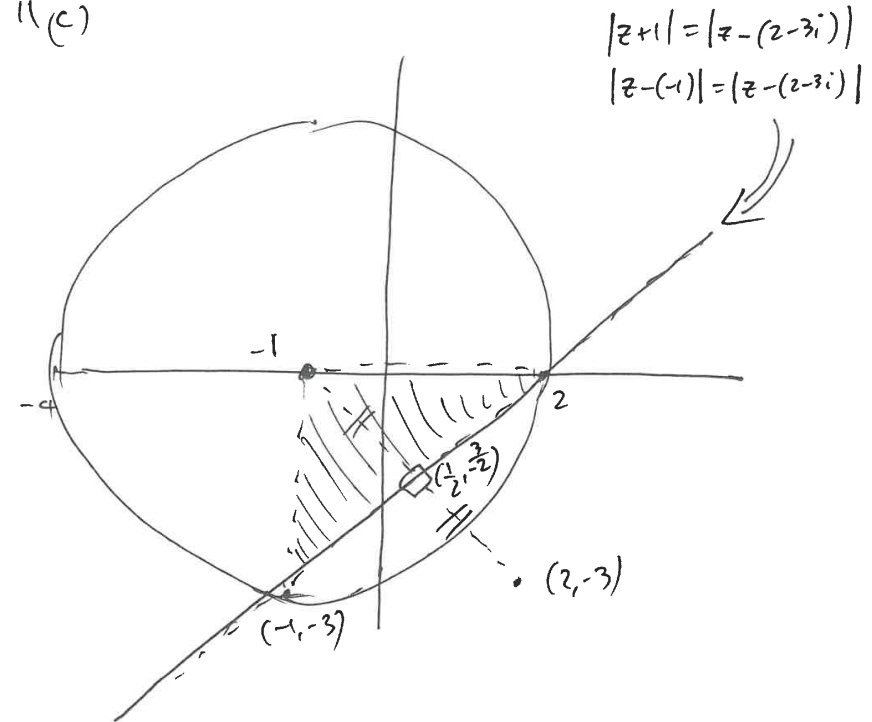
$$\leq \frac{k^4 + 4k^3 + 6k^2 + 4k + 1}{4}$$

$$= \frac{(k+1)^4}{4}$$

Hence  $\sum_{r=1}^{k+1} (r-1)^3 \leq \frac{(k+1)^4}{4}$ , as required.

Hence the statement is true by mathematical induction.

11(c)



So area of major segment =  $\frac{3}{4}$  of circle + shaded triangle  
 $= \frac{3}{4} \times \pi \times 2^2 + \frac{1}{2} \times 3 \times 3$   
 $= \frac{27\pi}{4} + \frac{9}{2} u^2$

Q17 (a) To prove that  $n^2+n$  is even for any integer  $n$ .

Case 1 Suppose  $n$  is even.

Let  $n = 2j$ .

$$\text{Then } n^2+n = (2j)^2+2j \\ = 4j^2+2j$$

$$= 2(2j^2+j), \text{ which is even.}$$

Case 2 Suppose  $n$  is odd.

Let  $n = 2k+1$ .

$$\text{Then } n^2+n = (2k+1)^2+(2k+1) \\ = 4k^2+4k+1+2k+1$$

$$= 4k^2+6k+2$$

$$= 2(2k^2+3k+1), \text{ which is even.}$$

$\therefore n^2+n$  is even for any integer  $n$ .

$$(b) i) (\cos \theta + i \sin \theta)^5 \\ = \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (i \sin \theta)^2 + 10 \cos^2 \theta (i \sin \theta)^3 + 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5 \\ = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta + i [5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta]$$

$$(ii) \text{ But } (\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

$$\text{So } \cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \\ \text{by equating Real parts}$$

$$\text{and } \sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta \\ \text{by equating imaginary parts}$$

$$\therefore \sin 5\theta = 5(1-\sin^2 \theta)^2 \sin \theta - 10(1-\sin^2 \theta) \sin^3 \theta + \sin^5 \theta \\ = 5 \sin \theta - 10 \sin^3 \theta + 5 \sin^5 \theta - 10 \sin^3 \theta + 10 \sin^5 \theta + \sin^5 \theta \\ = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta, \text{ as required.}$$

(iii) The solutions of  $16x^5 - 20x^3 + 5x - 1 = 0$

lie amongst the solutions to

$$\sin 5\theta = 1. \text{ if we let } x = \sin \theta$$

$$\text{Solving } \sin 5\theta = 1$$

$$5\theta = \left(\frac{\pi}{2} + 2k\pi\right), k=0,1,2,\dots$$

$$5\theta = \left(\frac{4k\pi + \pi}{2}\right)$$

$$\theta = \frac{4k\pi + \pi}{10}$$

$$\text{When } k=0, \theta = \frac{\pi}{10}.$$

Hence  $x = \sin \frac{\pi}{10}$  is one of the solutions of  $16x^5 - 20x^3 + 5x - 1 = 0$ .

$$(iv) \begin{array}{r} 16x^4 + 16x^3 - 4x^2 - 4x + 1 \\ (x-1) \overline{) 16x^5 + 0x^4 - 20x^3 + 0x^2 + 5x - 1} \\ \underline{16x^5 - 16x^4} \phantom{+ 0x^3 + 0x^2 + 5x - 1} \\ 16x^4 - 20x^3 \phantom{+ 0x^2 + 5x - 1} \\ \underline{16x^4 - 16x^3} \phantom{+ 0x^2 + 5x - 1} \\ -4x^3 + 0x^2 \phantom{+ 5x - 1} \\ \underline{-4x^3 + 4x^2} \phantom{+ 5x - 1} \\ -4x^2 + 5x \phantom{- 1} \\ \underline{-4x^2 + 4x} \phantom{- 1} \\ x - 1 \\ \underline{x - 1} \\ 0 \end{array}$$

$$\therefore P(x) = 16x^4 + 16x^3 - 4x^2 - 4x + 1$$

(v)  $(4x^2 + ax - 1)(4x^2 + ax - 1)$

The coefficient of the term in  $x^3$  is 16.

$$\therefore 4a + 4a = 16$$

$$a = 2$$

(vi) Clearly  $\sin \frac{\pi}{10}$  is a solution to  $P(x) = 0$

as  $\sin \frac{\pi}{10} \neq 1$ .

$$\text{Now } P(x) = (4x^2 + 2x - 1)^2$$

$$\therefore \text{Solve } 4x^2 + 2x - 1 = 0$$

$$x = \frac{-2 \pm \sqrt{4 + 16}}{8} = \frac{-2 \pm \sqrt{20}}{8} = \frac{-2 \pm 2\sqrt{5}}{8} = \frac{-1 \pm \sqrt{5}}{4}$$

$$\text{But } \sin \frac{\pi}{10} > 0, \therefore \sin \frac{\pi}{10} = \frac{\sqrt{5} - 1}{4}.$$

(vii) From (iii), other solutions than  $\sin \frac{\pi}{10}$  were:

$$k=1: \sin \frac{9\pi}{10} = \sin \frac{\pi}{2} = 1 \text{ which is the } (x-1) \text{ factor}$$

$$k=2: \sin \frac{7\pi}{10} = \sin \frac{\pi}{10} \times \text{duplicate}$$

$$k=3: \sin \frac{5\pi}{10} = -\sin \frac{3\pi}{10} = \frac{-\sqrt{5}-1}{4} \text{ (from (vi))}$$

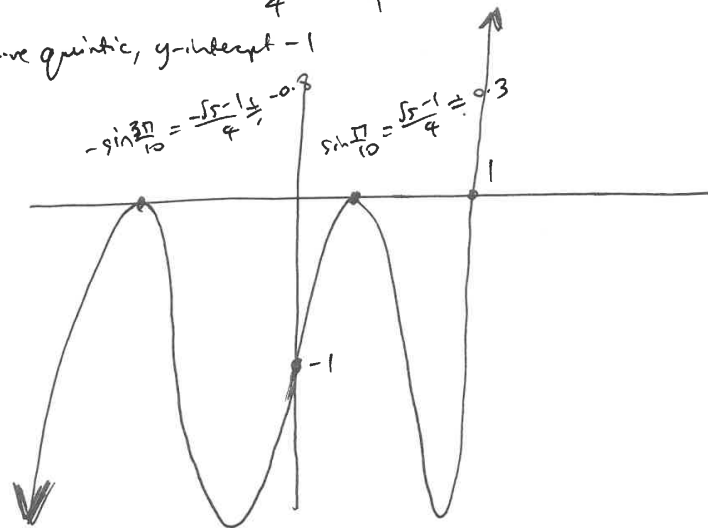
$$k=4: \sin \frac{3\pi}{10} = -\sin \frac{\pi}{10} \times \text{duplicate}$$

$$k=5: \sin \frac{\pi}{10} = \sin \frac{\pi}{10} \times \text{duplicate and around we go again.}$$

From (v) we have 2 double roots.

$$\therefore \text{roots are: } \frac{\sqrt{5}-1}{4}, \frac{\sqrt{5}-1}{4}, \frac{-\sqrt{5}-1}{4}, \frac{-\sqrt{5}-1}{4}, 1$$

Positive quintic, y-intercept -1



Q13 (a)

$$(i) 1 + \sqrt{3}i = 2e^{i\frac{\pi}{3}}$$

$$(ii) \text{ Now } z^4 = a(1 + \sqrt{3}i)$$

$$\therefore (2e^{i\frac{\pi}{3}})^4 = a(2e^{i\frac{\pi}{3}})$$

$$16e^{i\frac{4\pi}{3}} = 2ae^{i\frac{\pi}{3}}$$

$$\therefore a = 8$$

(iii) To find the other 3 roots:

$$z^4 = 8(2e^{i(\frac{\pi}{3} + 2k\pi)})$$

$$z^4 = 16e^{i(\frac{6k\pi + \pi}{12})}$$

$$\text{When } k=-1, -i\frac{5\pi}{12}$$

$$z_1 = 2e^{-i\frac{5\pi}{12}}$$

$$\text{When } k=1, i\frac{7\pi}{12}$$

$$z_2 = 2e^{i\frac{7\pi}{12}}$$

$$\text{When } k=2, i\frac{13\pi}{12}$$

$$z_3 = 2e^{i\frac{13\pi}{12}}$$

$$\text{ie, } z_3 = 2e^{-i\frac{11\pi}{12}}$$

(b) Let  $z = x + iy$  then

$$\frac{1}{x+iy} + \frac{1}{x-iy} = 1$$

$$\frac{2x}{x^2+y^2} = 1$$

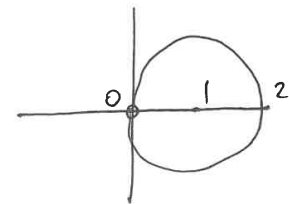
$$2x = x^2 + y^2$$

$$0 = x^2 - 2x + y^2$$

$$0 = (x-1)^2 + y^2 - 1$$

$$1 = (x-1)^2 + y^2$$

The locus of P is a circle,  
centre (1, 0) and radius = 1.



but a hole at  $z=0$  as

$\frac{1}{z}$  &  $\frac{1}{z}$  are  
undefined  
when  $z=0$

(c) Prove that

$$\frac{1}{a^2+1} + \frac{1}{b^2+1} \geq \frac{2}{ab+1} \quad \text{where } a, b > 0 \text{ and } ab \geq 1.$$

Consider LHS - RHS

$$= \frac{1}{a^2+1} + \frac{1}{b^2+1} - \frac{2}{ab+1}$$

$$= \frac{(a^2+b^2+2)(ab+1) - 2(a^2+1)(b^2+1)}{(a^2+1)(b^2+1)(ab+1)}$$

$$= \frac{a^3b + ab^3 + 2ab + a^2 + b^2 + 2 - 2a^2b^2 - 2a^2 - 2b^2 - 2}{(a^2+1)(b^2+1)(ab+1)}$$

$$= \frac{a^3b + ab^3 - a^2 - b^2 + 2ab - 2a^2b^2}{(a^2+1)(b^2+1)(ab+1)}$$

$$= \frac{ab(a^2+b^2) - 1(a^2+b^2) + 2ab(1-ab)}{(a^2+1)(b^2+1)(ab+1)}$$

$$= \frac{(ab-1)(a^2+b^2) - 2ab(ab-1)}{(a^2+1)(b^2+1)(ab+1)}$$

$$= \frac{(ab-1)(a^2-2ab+b^2)}{(a^2+1)(b^2+1)(ab+1)}$$

$$= \frac{(ab-1)(a-b)^2}{(a^2+1)(b^2+1)(ab+1)}$$

which equals 0 when  $a=b$ ,  
but is otherwise positive  
since  $ab \geq 1$ .

Hence LHS  $\geq$  RHS.

Q14 (a)  $z$  is one of the fifth roots of unity other than 1. So  $z$  is

one of

$$\text{cis}\left(\frac{2\pi}{5}\right), \text{cis}\left(\frac{4\pi}{5}\right), \text{cis}\left(\frac{6\pi}{5}\right), \text{cis}\left(\frac{8\pi}{5}\right).$$

Let  $z = \text{cis}\left(\frac{2\pi}{5}\right)$  then

$$z - z^2 + z^3 - z^4$$

$$= \text{cis}\left(\frac{2\pi}{5}\right) - \text{cis}\left(\frac{4\pi}{5}\right) + \text{cis}\left(\frac{6\pi}{5}\right) - \text{cis}\left(\frac{8\pi}{5}\right)$$

$$= \cos\left(\frac{2\pi}{5}\right) + i\sin\left(\frac{2\pi}{5}\right) - [\cos\left(\frac{4\pi}{5}\right) + i\sin\left(\frac{4\pi}{5}\right)] + [-\cos\left(\frac{6\pi}{5}\right) - i\sin\left(\frac{6\pi}{5}\right)] - [-\cos\left(\frac{8\pi}{5}\right) + i\sin\left(\frac{8\pi}{5}\right)]$$

$$= \cos\left(\frac{2\pi}{5}\right) + i\sin\left(\frac{2\pi}{5}\right) - \cos\left(\frac{4\pi}{5}\right) - i\sin\left(\frac{4\pi}{5}\right) - \cos\left(\frac{6\pi}{5}\right) - i\sin\left(\frac{6\pi}{5}\right) + \cos\left(\frac{8\pi}{5}\right) - i\sin\left(\frac{8\pi}{5}\right)$$

$$= 2i \sin\left(\frac{2\pi}{5}\right) - 2i \sin\left(\frac{4\pi}{5}\right)$$

$$= 2i \left( \sin\frac{2\pi}{5} - \sin\frac{4\pi}{5} \right), \text{ as required.}$$

(b) (i) Consider RHS - LHS

$$= \frac{1}{a-1} - \frac{1}{a} - \frac{1}{a^2}$$

$$= \frac{a^2 - a(a-1) - (a-1)}{a^2(a-1)}$$

$$= \frac{a^2 - a^2 + a - a + 1}{a^2(a-1)}$$

$$= \frac{1}{a^2(a-1)}, \text{ which is a positive expression as } a \geq 2.$$

Hence RHS  $>$  LHS.

$$\therefore \frac{1}{a^2} < \frac{1}{a-1} - \frac{1}{a}$$

$$\begin{aligned}
 \text{(ii)} \quad & \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots + \frac{1}{n^2} \\
 & < \frac{1}{1^2} + \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n-2} - \frac{1}{n-1} + \frac{1}{n-1} - \frac{1}{n} \\
 & = 2 - \frac{1}{n} \\
 & < 2 \text{ for positive integer } n \\
 \text{Hence } & \frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{n^2} < 2
 \end{aligned}$$

$$\text{(c) (i)} \quad f(x) = e^x - x$$

$$f'(x) = e^x - 1$$

$$f'(0) = 0$$

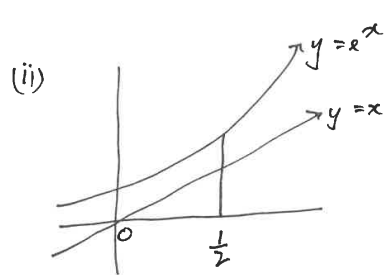
$\therefore (0, 1)$  is a stationary point

$$f''(x) = e^x$$

$$f''(0) = 1 > 0$$

$\therefore (0, 1)$  is a minimum turning point.

$\therefore f(x) = e^x - x$  is increasing for all  $x > 0$   
 $\therefore$  for  $x > 0, f(x) > 1 \therefore e^x - x > 1 \therefore e^x > x + 1$   
 $\therefore e^x > x$  for all  $x \geq 0$ , noting that  $e^0 = 1$



noting from (i) that  
 $e^x > x$  for  $x \geq 0$

$$\text{Now } \int_0^{\frac{1}{2}} e^x dx > \int_0^{\frac{1}{2}} x dx$$

$$[e^x]_0^{\frac{1}{2}} > \left[\frac{x^2}{2}\right]_0^{\frac{1}{2}}$$

$$\sqrt{e} - 1 > \frac{1}{8}$$

$$\therefore \sqrt{e} > \frac{1}{8}$$

$$\therefore \sqrt{e} > \frac{9}{8} \text{ as required.}$$

(iii) To prove that  $e^x > \frac{x^n}{n!}$  for all integers  $n \geq 1, x \geq 0$

$$\text{When } n=1, \text{ LHS} = e^x$$

$$\text{RHS} = \frac{x^1}{1!} = x$$

Since  $e^x > x$  from part (i) then  $\text{LHS} > \text{RHS}$ .

So the statement is true when  $n=1$ .

Let the statement be true when  $n=k$ , i.e., let

$$e^x > \frac{x^k}{k!} \text{ for } x \geq 0$$

and then try to prove the statement is true for  $n=k+1$ , i.e.,

$$\text{try to prove that } e^x > \frac{x^{k+1}}{(k+1)!} \text{ for } x \geq 0$$

$$\text{Given } e^x > \frac{x^k}{k!}$$

$$\text{then } \int_0^t e^x dx > \int_0^t \frac{x^k}{k!} dx$$

$$[e^x]_0^t > \left[\frac{x^{k+1}}{(k+1)!}\right]_0^t$$

$$e^t - 1 > \frac{t^{k+1}}{(k+1)!} - 0$$

$$\therefore e^t > 1 + \frac{t^{k+1}}{(k+1)!}$$

$$\therefore e^t > \frac{t^{k+1}}{(k+1)!}$$

$$\therefore e^x > \frac{x^{k+1}}{(k+1)!} \text{ as required.} \quad \#$$